

The Little Big Data Revolution

BIG DATA, THE NAME ALMOST HAS A “BIG BROTHER” VIBE TO IT. A massive cache of information about anything and everything pertaining to your project, stored in massive storage clusters which have been designed to handle petabytes upon petabytes. Big Data was and, to a certain extent, does provide users with key insights or, as they call it, business intelligence thanks to a tonne of data points to sift through. However, you could say Big Data was a little ahead of its time. The method of ingesting data into unstructured databases also presented with the problem of figuring out how one could go about extracting the right kind of insights. We’re talking petabytes of data and with each company having to customise the data models, you had a massive amount of fragmentation in the market in terms of the kind of approaches that one could take to tackle a need. There would be barely any fast methods of extracting insights since each company would go about it in a slightly different manner. Only the big tech companies could afford to have large teams to optimise the software and hardware for big data. Otherwise, you were looking at getting query results quite late. And there’s no point in holding onto obsolete information, at least not at the sake of the ridiculous expenditure involved in propping up these systems.

Whether you used Qlik, Tableau or any of the hundreds of other solutions out there, you wouldn’t get the insights that you needed at the right time. This led to a spur in innovation as faster storage mediums became the need of the hour. SSDs on PCIe busses that were a much hotter form of persistent storage became necessary because moving data from the traditional SSD to RAM before running data analytics would entail a significant waiting period as terabytes of relevant data had to be transferred for each query. Being able to run queries on a fast storage where data doesn’t have to move around is much preferred over methods that require a lot of movement. Speeding up data queries would entail minimal data movement, proper structure and reduced redundant data.

A typical process for extracting tangible insights takes weeks and the need of the

hour is to extract the same within hours, if not within minutes. A huge reason for such long wait times is data blending which basically means merging data from multiple sources. For this, we’ve seen tiered analytics being put in place. Expected queries are worked in the background as soon as the data is ingested so that when the demand arises, you could provide the insights right away. One good example would be how Google Analytics has AI models running to provide proper insights on the basis of the typical queries presented to it by the hundreds of thousands of individuals and companies that use the service. However, Google Analytics is very structured and it’s easy for them to extract insights within seconds. The rest of the world is more concerned with unstructured data since that’s what they typically have to work with.

While enterprises will continue to stick with their custom big data solutions, smaller big data applications can leverage GPUs. The recent launches from NVIDIA and AMD have insanely massive onboard memory caches. Take the latest Quadro RTX 8000, it has 48 GB of onboard storage but it’s a card designed for creative workflows that can easily be programmed for big data applications. And thanks to new technologies such as Microsoft’s DirectStorage, you can issue a lot more I/O calls to the NVMe SSDs that are connected and pull data from SSD to VRAM super fast. And if you happen to have a PCIe Gen 4.0 system, then you can move data even faster. It would be interesting to see how such applications could be put to use in companies that don’t have a lot of money to throw around. Setting up a cloud based big data solution is a recurring expense while an on-prem GPU setup would be a simple one time expense. In a way, we could see Big Data applications getting democratised and see it becoming more commonplace than what it is today. Turns out that the new graphics cards aren’t just exciting for gamers. **d**



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